

Çukurova Üniversitesi, Bilgisayar Mühendisliği

DIGITAL DESIGN Midterm Exam (2025 Çıkmış Sorular)

(Sedat Bilgili, 2025)

1. (10p) 5 questions on converting numbers between the binary (including two's complement, negatives) and decimal systems.

2. (30p) Designing a logic system from a given "story" or scenario:

- Creating the Truth Table.
- Boolean Equation Simplification (Using K-Maps).
- Minterm / Maxterm Notation: Expressing the function in Sum-of-Products (Minterm) or Product-of-Sums (Maxterm) form.
- Circuit Diagram: Drawing the final, simplified logic circuit.

(B,P,S,L input Y output | L = 0 iken S ya da P = 1 ise Y=1 | L=1 iken S=1 ise Y=1 | B=0 ise her zaman Y=0)

3. (10p) Explain how a 4-to-1 MUX operates:

- Draw its Truth Table.
- Draw its internal logic circuit diagram.

4. (10p) Explaining the fundamental differences between Moore and Mealy FSMs.

5. (10p) Timing Analysis: Calculating signal delay (e.g., propagation delay) in a given circuit based on the timing of input value changes.

6. (30p) Design a Moore FSM that detects the sequence "101" from a serial input (X):

- Define the states.
- Create the State Diagram.
- Create the State Transition Table and Output Table.
- Perform State Encoding and Output Encoding.
- Draw the general logic circuit for the complete Moore machine (using flip-flops and gates).